EduFin SQL Business Challenge

Phase 2 — Tuesday Submission

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STEP 2A — City-Level Loan Volume

```
SELECT
    dc.city_name AS `City`,
    COUNT(*) AS `Total Loans`
FROM loans l
INNER JOIN customers c ON l.customer_id = c.customer_id
INNER JOIN dim_city dc ON c.city_id = dc.city_id
GROUP BY dc.city_name
ORDER BY `Total Loans` DESC;
```

```
+-----------+-----------+
|       City|Total Loans|
+-----------+-----------+
|      Delhi|        209|
|     Raipur|        207|
|     Bhopal|        204|
|Bhubaneswar|        199|
|    Chennai|        191|
|      Patna|        183|
|    Lucknow|        179|
|     Nagpur|        176|
|   Guwahati|        176|
|     Indore|        170|
+-----------+-----------+
only showing top 10 rows
```

STEP 2B — Default Count and Rate Per City

```
SELECT
    dc.city_name AS `City`,
    COUNT(*) AS `Total Loans`,
    SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) AS `Total Defaults`,
    CONCAT(
        ROUND(
            (SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100.0 /
COUNT(*)),
            2
        ),
        '%'
    ) AS `Default Rate (%)`
FROM loans l
INNER JOIN customers c ON l.customer_id = c.customer_id
INNER JOIN dim_city dc ON c.city_id = dc.city_id
GROUP BY dc.city_name
```

```
HAVING COUNT(*) >= 100
ORDER BY CAST(REPLACE(`Default Rate (%)`, '%', '') AS DOUBLE) DESC;
```

City	Total Loans	Total Defaults	Default Rate (%)
Pune	149	30	20.13%
Ranchi	168	30	17.86%
Indore	170	28	16.47%
Jaipur	160	26	16.25%
Rajkot	166	26	15.66%
Kochi	165	25	15.15%
Ahmedabad	160	24	15.00%
Jammu	159	23	14.47%
Nashik	135	19	14.07%
Lucknow	179	25	13.97%

only showing top 10 rows

STEP 2C — Financial Exposure Per City

```
SELECT
    dc.city_name AS `City`,
    COUNT(*) AS `Total Loans`,
    CONCAT(ROUND(SUM(l.loan_amount) / 10000000.0, 2), ' Cr') AS `Total Portfolio
Values`,
    CASE
        WHEN SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0 ELSE 0
END) > 10000000.0
        THEN CONCAT(ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount
* 1.0 ELSE 0 END) / 10000000.0, 2), ' Cr')
        ELSE CONCAT(ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount
* 1.0 ELSE 0 END) / 100000.0, 2), ' L')
    END AS `Total Defaulted Values`
FROM loans l
INNER JOIN customers c ON l.customer_id = c.customer_id
INNER JOIN dim_city dc ON c.city_id = dc.city_id
GROUP BY dc.city_name
ORDER BY SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0 ELSE 0
END) DESC;
```

```
+---------+-----------+----------------------+----------------------+
|     City|Total Loans|Total Portfolio Values|Total Defaulted Values|
+---------+-----------+----------------------+----------------------+
|   Ranchi|        168|               6.71 Cr|                1.4 Cr|
|     Pune|        149|               5.87 Cr|               1.14 Cr|
|  Lucknow|        179|               7.51 Cr|               1.13 Cr|
|   Mumbai|        168|               6.59 Cr|               1.06 Cr|
|Ahmedabad|        160|               7.07 Cr|               1.05 Cr|
|   Indore|        170|               6.73 Cr|               1.04 Cr|
|   Jaipur|        160|               6.58 Cr|               1.03 Cr|
|    Kochi|        165|               6.67 Cr|                1.0 Cr|
|    Jammu|        159|               6.52 Cr|              98.62 L|
|   Rajkot|        166|               6.32 Cr|              97.93 L|
|   Bhopal|        204|               8.53 Cr|              84.25 L|
|   Nashik|        135|               5.62 Cr|              83.66 L|
|    Patna|        183|               7.77 Cr|              83.61 L|
|    Delhi|        209|               8.22 Cr|              83.12 L|
|   Nagpur|        176|               7.06 Cr|              82.25 L|
+---------+-----------+----------------------+----------------------+
only showing top 15 rows
```

STEP 2D — Geographic Risk Pattern (State-Level Rollup)

```
SELECT
    dc.city_name AS `City`,
    CONCAT(ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100 /
COUNT(*), 2), '%') AS `Default Rate (%)`,
    CASE
        WHEN ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100
/ COUNT(*), 2) > 15 THEN 'Critical'
        WHEN ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100
/ COUNT(*), 2) > 10 THEN 'High Risk'
        WHEN ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100
/ COUNT(*), 2) > 5 THEN 'Monitor'
        ELSE 'Safe'
    END AS `Risk Category`
FROM loans l
INNER JOIN customers c ON l.customer_id = c.customer_id
INNER JOIN dim_city dc ON c.city_id = dc.city_id
GROUP BY dc.city_name
HAVING `Risk Category` IN ('Critical', 'High Risk')
ORDER BY CAST(REPLACE(`Default Rate (%)`, '%', '') AS DOUBLE) DESC;
```

```
+---------+----------------+-------------+
|     City|Default Rate (%)|Risk Category|
+---------+----------------+-------------+
|     Pune|          20.13%|     Critical|
|   Ranchi|          17.86%|     Critical|
|   Indore|          16.47%|     Critical|
|   Jaipur|          16.25%|     Critical|
|   Rajkot|          15.66%|     Critical|
|    Kochi|          15.15%|     Critical|
|Ahmedabad|           15.0%|    High Risk|
|    Jammu|          14.47%|    High Risk|
|   Nashik|          14.07%|    High Risk|
|  Lucknow|          13.97%|    High Risk|
+---------+----------------+-------------+
only showing top 10 rows
```

STEP 2E — Regional Crisis Dashboard (Final Query)

```
SELECT
    dc.city_name AS `City`,
    COUNT(*) AS `Loans`,
    CONCAT(ROUND(SUM(l.loan_amount) / 10000000.0, 2), ' Cr') AS `Portfolio Value`,
    CASE
        WHEN SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0 ELSE 0
END) > 10000000.0
        THEN CONCAT(ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount
* 1.0 ELSE 0 END) / 10000000.0, 2), ' Cr')
        ELSE CONCAT(ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount
* 1.0 ELSE 0 END) / 100000.0, 2), ' L')
    END AS `Total Defaulted Values`,
    CONCAT(ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100 /
COUNT(*), 2), '%') AS `Default Rate`,
    ROUND(
        (ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100.0 /
COUNT(*), 2)) *
        (ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0 ELSE
0 END) / 10000000.0, 2)),
        2
    ) AS `Impact Score`,
    CASE
        WHEN ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100
/ COUNT(*), 2) > 15 THEN 'Critical'
        WHEN ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100
/ COUNT(*), 2) > 10 THEN 'High Risk'
        WHEN ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN 1 ELSE 0 END) * 100
/ COUNT(*), 2) > 5 THEN 'Monitor'
        ELSE 'Safe'
    END AS `Risk Classification`
FROM loans l
INNER JOIN customers c ON l.customer_id = c.customer_id
INNER JOIN dim_city dc ON c.city_id = dc.city_id
GROUP BY dc.city_name
HAVING COUNT(*) >= 100
ORDER BY ROUND(SUM(CASE WHEN l.loan_status = 'Defaulted' THEN l.loan_amount * 1.0 ELSE
0 END) / 10000000.0, 2) DESC;
```

```
+----------+-----+--------------+---------------------+------------+------------+------------------+
|      City|Loans|Portfolio Value|Total Defaulted Values|Default Rate|Impact Score|Risk Classification|
+----------+-----+--------------+---------------------+------------+------------+------------------+
|    Ranchi|  168|        6.71 Cr|               1.4 Cr|      17.86%|        25.0|          Critical|
|      Pune|  149|        5.87 Cr|              1.14 Cr|      20.13%|       22.95|          Critical|
|   Lucknow|  179|        7.51 Cr|              1.13 Cr|      13.97%|       15.79|         High Risk|
|    Mumbai|  168|        6.59 Cr|              1.06 Cr|       13.1%|       13.89|         High Risk|
| Ahmedabad|  160|        7.07 Cr|              1.05 Cr|       15.0%|       15.75|         High Risk|
|    Indore|  170|        6.73 Cr|              1.04 Cr|      16.47%|       17.13|          Critical|
|    Jaipur|  160|        6.58 Cr|              1.03 Cr|      16.25%|       16.74|          Critical|
|     Kochi|  165|        6.67 Cr|               1.0 Cr|      15.15%|       15.15|          Critical|
|     Jammu|  159|        6.52 Cr|             98.62 L|      14.47%|       14.33|         High Risk|
|    Rajkot|  166|        6.32 Cr|             97.93 L|      15.66%|       15.35|          Critical|
|     Patna|  183|        7.77 Cr|             83.61 L|       9.29%|         7.8|           Monitor|
|    Bhopal|  204|        8.53 Cr|             84.25 L|      10.78%|        9.06|         High Risk|
|    Nashik|  135|        5.62 Cr|             83.66 L|      14.07%|       11.82|         High Risk|
|     Delhi|  209|        8.22 Cr|             83.12 L|      10.53%|        8.74|         High Risk|
|    Nagpur|  176|        7.06 Cr|             82.25 L|       12.5%|       10.25|         High Risk|
|    Raipur|  207|         8.5 Cr|             79.77 L|      10.14%|        8.11|         High Risk|
|    Shimla|  144|        5.99 Cr|             75.95 L|      13.19%|       10.02|         High Risk|
|   Gangtok|  150|        6.08 Cr|             75.38 L|      11.33%|         8.5|         High Risk|
|Chandigarh|  168|        7.09 Cr|              75.3 L|      10.71%|        8.03|         High Risk|
| Hyderabad|  159|         6.7 Cr|             72.95 L|      10.06%|        7.34|         High Risk|
+----------+-----+--------------+---------------------+------------+------------+------------------+
only showing top 20 rows
```

```
The 3 cities to halt lending it immediately are Ranchi, Pune and Indore due to their high default values and rate reaching
upto crtitcal impact.
```

Custom Column

```
RANK() OVER (
    ORDER BY (default_rate * default_exposure) DESC
) AS exposure_rank

CASE NTILE(4) OVER (ORDER BY default_exposure DESC)
   WHEN 1 THEN 'Critical'
   WHEN 2 THEN 'High'
   WHEN 3 THEN 'Moderate'
   ELSE 'Low'
END AS risk_quartile
```

> NTILE is better than hardcoded CASE thresholds because it automatically splits the cities into equal sized groups (quartiles) and also automatically adapts when new cities are added. Also since our portfolio data can grow or shrink in the near future, the hardcoded thresholds may become outdated making the user update the query time and again. However, in cases where you must stay consistent due to company policies, it is mandatory to use CASE thresholds as they can strictly follow absolute business rules.

WHY Gate

1. If you sorted your final list by default rate instead of exposure_rank — does your top-3 halt recommendation change? Name the specific cities that move in or out, and explain what that difference means for the CEO's decision.

> The top 3 halt recommendation changes from ('Ranchi', 'Pune', 'Indore') using exposure_rank to ('Pune', 'Ranchi', 'Indore') using default_rank, meaning that no city moved in or out and only the order between 'Ranchi' and 'Pune' has been switched. This difference highlights that default rate prioritizes the frequency of loan failures, making 'Pune' appear riskier due to a higher percentage

of defaults. In contrast, exposure-based ranking prioritizes financial impact, where 'Ranchi' ranks higher because it has a larger amount of capital at risk.

2. You excluded cities with fewer than 100 loans. Name one city that was excluded and explain: why does excluding it protect the business from making a bad decision?

No, city was excluded as there were no cities with fewer than 100 loans.

3. Write your 3-city halt decision memo. Format: City Name → default rate → defaulted exposure → specific reason to halt THIS city and not the next one on the list.

- 'Ranchi' → 17.86% → 1.4 Cr → It is the highest financial exposure in the portfolio (1.40 Cr), combined with one of the highest default rates. - 'Pune' → 20.13% → 1.14 Cr → It has the highest default rate (20.13%) from our entire list, while also having a decent financial exposure, making it the city with the worst failure frequency. - 'Indore' → 16.47% → 1.04 Cr → While exposure is slightly lower than Pune and Ranchi, both default rate and absolute loss remain high enough to place it in the critical zone.

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